

Amoeba Sisters Video Recap: Osmosis

1. The below picture represents diffusion of molecules. Place the following labels in the diagram: high concentration, low concentration, and an arrow showing the direction that the molecules would travel before equilibrium is reached.



2. Osmosis is a type of diffusion, but it involves the movement of water. Similar to diffusion, osmosis is the movement of molecules (water molecules if osmosis) from a high concentration to a low concentration.

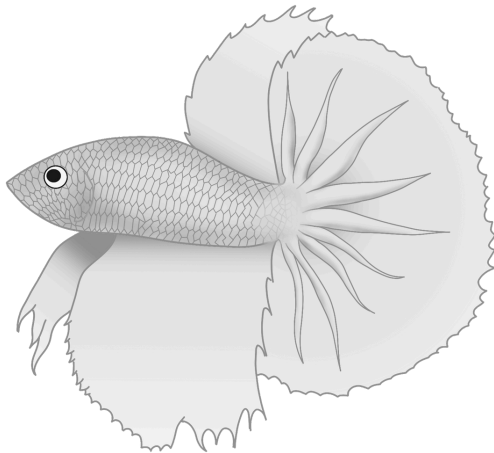
The video clip explains that you can also look at water as moving to a _____ concentration of solute molecules.

Why can it also be viewed this way?

3. Osmosis Scenario: The video clip mentioned a disaster scenario of a saltwater fish being placed in fresh water.

What would occur if, instead, a freshwater fish was placed in saltwater?

Your answer needs to have an arrow indicating the direction of water flow in osmosis, a label for "hypertonic," and a label for "hypotonic."



4. Osmosis Scenario: Fluid movement into the brain after traumatic brain injury can result in dangerous brain swelling.

One treatment that can be used in some of these cases is adding a _____ saline. You need to decide whether this blank should be the word hypertonic or hypotonic. Remember, you are trying to reduce the excessive fluid in the brain.

Explain your answer:



The Gummy Bear Mystery

The gummy bears are greatly enlarged by the time you get home! Your sister and some friends have different viewpoints for what happened. Draw a diagram below showing the enlarged gummy bears in a cup of water. Place the labels “hypertonic” and “hypotonic” in your diagram. One label should be for the gummy bears and one label should be for the water.

[illegible]

Stable